

Institutional analysis of evolution of joint forest management in India: A new institutional economics approach

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Abstract

This paper analyzes the evolution of the Joint Forest Management (JFM) institution in India in the light of the New Institutional Economics. To structure the analysis and shed light on different sources of persistent problems related to JFM, the paper uses the ‘four levels of institutional analysis’ framework developed by Williamson. In this paper we focus particularly on level 2 and 3. The second level of institutional analysis refers to the institutional environment (‘formal rules of the game’). We argue that the shift from state to co-management is a step in the right direction given the realities encountered in India. The insecure, incomplete and often incoherent transfer of specific rights from the state to local communities is, however, an important source of incentive problems on the part of communities. The third level of institutional analysis deals with the governance of contractual relations (‘play of the game’). Here, the interactions between forest departments and communities are discussed at a more disaggregate level. It is shown that information asymmetries and lack of accountability lead to a prevalence of enforcement problems and rent-seeking activities, putting into question the efficiency of the current system. We conclude with policy recommendations and issues for future research.

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1. Introduction

Over the last two decades, devolution of resource management and access rights from the state to local communities and user groups has become an important policy tool in developing countries. The apparent change in policy from the traditional state-managed

top-down approach to the community level is fueled by the recognition of the limits of government agencies in managing resources at the local level, which have resulted in massive degradation of natural resources and local livelihood systems.

India has been at the forefront of devolving natural resource management to the local-community level, particularly in the forestry sector, for more than a decade. Its 1988 forest policy was a landmark, because for the first time it recognized the importance of local people’s involvement in forest management for achieving improvements in community livelihood

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and the protection of forest resources. In a follow-up document issued in 1990, the central government issued guidelines to all state governments to implement ‘Joint Forest Management Systems’ by transferring everyday forest use and management rights to the community. Accordingly, all the states have formally resolved to implement JFM, making it one of the largest of such programs in the world (Kumar, 2002).

Empirical evidence on the outcomes of JFM is mixed. Some studies have shown improvements in outcomes such as increased yields of timber and non-timber forest products (NTFPs), fuelwood, and fodder (Joshi, 1999; Khare et al., 2000; Ballabh et al., 2002). Others indicate a lack of effective control and management of forests by communities (Sundar, 2001; Bulte and Engel, *in press*). It is also argued that the differences in outcomes of JFM may partly be due to the fact that the degree to which specific rights and benefits were actually devolved from the forest department to the local communities differs significantly across states (Damodaran and Engel, 2003; Sarin, 2002; Ballabh et al., 2002; Agrawal and Ostrom, 2002).

The objective of this paper is to analyze the evolution of the Joint Forest Management (JFM) institution in India and sources of persistent problems in the light of the ‘New Institutional Economics’. To structure the analysis the paper uses the ‘four levels of institutional analysis’ of Williamson (2000). To limit the scope of the paper, we focus particularly on level 2 (‘formal rules of the game’) and 3 (‘play of the game’), but levels 1 and 4 are briefly reviewed and their importance highlighted. The paper is based on a review of existing literature and complemented with results from a recent field study conducted by the authors in 55 JFM villages in the state of Andhra Pradesh.

The remainder of this paper is organized as follows. Section 2 briefly reviews Williamson’s framework and presents information on the field study. Section 3 discusses the formal institutional environment (level 2), based on a detailed analysis of the evolution of property rights under JFM. Section 4 analyzes the interactions and contractual obligations between Forest Departments (FDs) and communities at a more disaggregate level (level 3). Issues of rule setting, monitoring and enforcement within JFM institutions are discussed. Section 5 concludes.

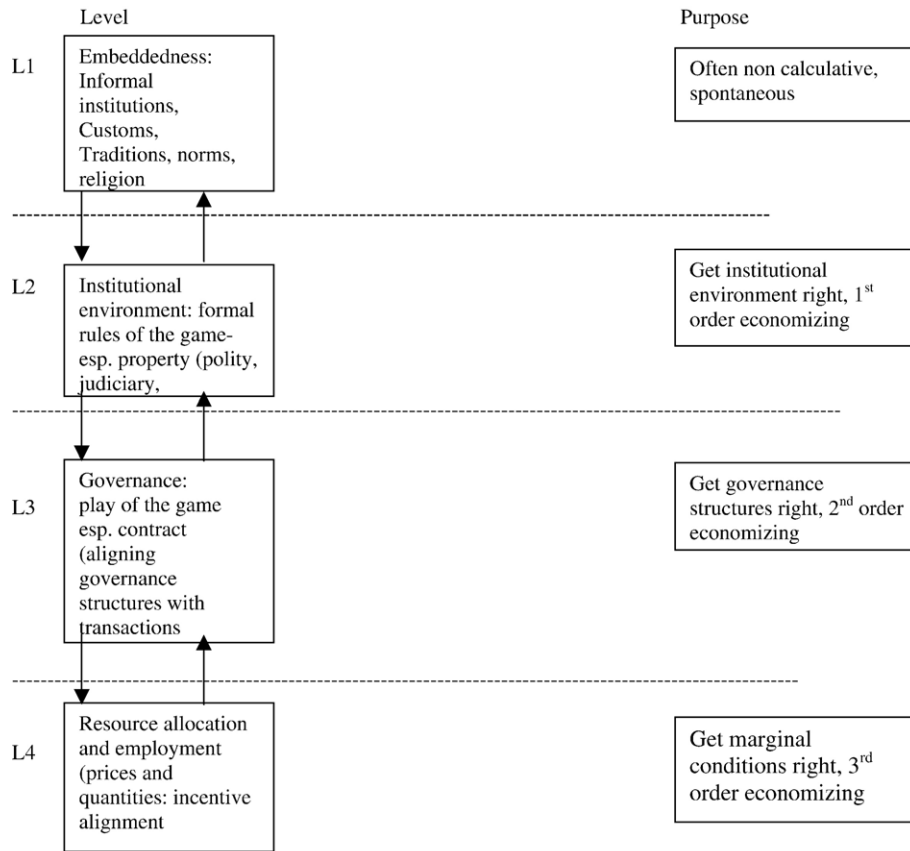
2. Background

2.1. Four levels of institutional analysis

To identify the complex and manifold influences that an economic institution such as JFM is subjected to demands an analytical structure. Williamson (2000) provides such a structure to analyze complex institutions. Analyzing the evolution and functioning of JFM in Williamson’s framework helps to shed light on the sources of problems experienced under JFM and to gain insights on how to overcome them.

Specifically, Williamson (2000) distinguishes four stages in which an institution functions (Fig. 1). The first level, ‘embeddedness’, deals with informal constraints (e.g., customs, taboos, traditional norms) that the institution is embedded in. Such informal constraints are usually present in a society for a long period, and are believed to be spontaneous in origin. The second level refers to the institutional environment that human beings create through formal rules (e.g., constitutions, laws, property rights, etc.). Institutions at this level tend to change faster than at the previous one. The theoretical framework for this level is given by the economics of property rights. Level 3 deals with how the game is played or ‘the governance of contractual relations’. Here, institutions of governance are discussed, including issues of contract definition and enforcement, as well as conflict resolution mechanisms. Changes at this level take place within shorter time than at the previous levels. Level 4 deals with economic activities and is concerned with getting the marginal conditions right. At this level the incentives are shaped by the combined effects of the other three levels.

Stages are interconnected with each other. In particular, higher level institutions impose constraints on lower levels. There may also be feedback effects from lower to higher levels. To limit the scope of this paper, we focus on levels 2 and 3 of the analysis. Levels 2 and 3 together provide a complete analysis of property rights, the core of the new institutional economics (NIE). In particular, the second level relates to absolute property rights, i.e., property rights in physical objects that guarantee the owner a power that he can exercise against all others (Furubotn and Richter, 2000, p. 75). The third



Source: Williamson (2000)

Fig. 1. Economics of institutions.

level is primarily concerned with relative property rights or contractual rights which deal with issues relating to contracts and their enforcement (Furubotn and Richter, 2000). In particular, we will discuss the actual interactions between local communities and forest departments in terms of rule setting and enforcement.

Nevertheless, it should be stressed that level 1 and 4 are clearly important in influencing the evolution of institutions in general and JFM in particular. Informal constraints such as norms, culture, conventions, and codes of ethics (level 1) evolve over time in a society and are found to be more pervasive than formal rules. These informal institutions do not change immediately in reaction to changes in formal rules. As a result, they play an important role in structuring inter-

actions among the users of forest resources. Tensions between changed formal rules and persisting informal institutions have implications for overall socio-economic outcomes (North, 1990). Such informal institutions are clearly present and of great importance in India, but a detailed discussion of these and the associated concept of culture would have to be vast and is beyond the scope of this paper. Similarly, level 4 deals with household-level incentives and the participation of individual households in community activities. Important issues arise here, particularly with respect to (i) who actually participates in community-level decision-making, and (ii) the intra-community distribution of benefits from JFM. At this level the opportunity costs of participation of FPC members in the JFM program are determined and influence the out-

comes. Again, we leave this important topic for future research.

2.2. Field study

This paper is based on a review of the existing literature as well as results from a recent field study conducted by the authors in the state of Andhra Pradesh during the period from September 2003 to February 2004. Andhra Pradesh is the second largest JFM state in terms of the number of Forest Protection Committees in India. The villages were selected using ‘purposive random sampling’ in order to ensure a mixed sample of well-performing and non-performing JFM villages. Sampling was based on FD performance ratings. A total of 147 JFM villages were visited and informal discussions held with executive and general body members, and the results were used to update the FD statistics. Finally, 55 JFM villages were selected for detailed analysis. Wealth ranking was carried out in each of these villages in order to gather basic information on caste, land and livestock holding of all village households and to classify households into poor, middle and rich on the basis of their land holdings.¹ A sample of 12 households was selected from each village, with 4 households from each of the above wealth categories. In order to make results representative at the level of the sample frame all village and household-level analysis is conducted using weights, which are inversely related to the likelihood of each village and household being sampled (Deaton, 1997).

Both quantitative and qualitative information on the details of forest use and management and the functioning of FPCs were gathered. Two structured questionnaires were prepared in order to elicit information at the village and household levels. At the village level interviews with groups of key informants were conducted. Household heads were interviewed to gather information at the household level. In addition, qualitative methods such as focus group discussions, transect walks, discussions with local Forest Department officials and state and local NGOs were applied.

3. Institutional environment (formal rules of the game)

The manner in which people use environmental resources depends on the property rights governing these resources (Tietenberg, 1994). In particular, property rights significantly affect the incentives individuals face, which ultimately determine the final outcome of resource management. An important reason for the massive degradation of natural resources in developing countries is a lack of well-defined and secure property rights (Panayotou, 1993; Pearce and Warford, 1993). Since Hardin’s (1968) famous article on ‘The tragedy of the commons’, there has been substantial debate on the most appropriate property right regime for common-pool resources such as forests. According to the property rights school, privatization of natural resources among potential users will lead to efficient use and allocation through market forces as it provides resource owners with incentives to undertake required investments to improve resource conditions (Demsetz, 1967). However, it is also believed that private property may not lead to efficient use and management of resources where the costs of exclusion are very high, which may lead to difficulties in making a complete assignment of property rights to individuals (Furubotn and Richter, 2000). This is true particularly in the case of use and management of natural resources in developing countries where many users depend on these resources for their livelihoods. In such a situation the potential of common ownership of natural resources along with self-crafted rules by the users on how to use the resources has been proposed as a potential solution (Ostrom, 1990). More recently, the benefits of a co-management regime, where both the state and local communities have some rights and responsibilities over the resource, are increasingly realized (e.g., Baland and Platteau, 1996).

In this section, we first describe and assess the shift from a state property regime to a co-management regime that has occurred in India. We then proceed to analyze in detail to what degree various types of rights have been transferred to FPCs.

3.1. From state management to co-management

The property rights regime over forest resources in India has been undergoing considerable change over

¹ The classifications are done in the range of 0–3 ha, 3–5 ha, and >5 ha for poor, middle, and rich, respectively.

time. During the pre-colonial period forests were controlled and managed by village communities, resulting in a common property regime with no private claims by individuals, and forest access for all community members according to their needs (Ghate, 2003). This property regime changed when the first forest policy statement of the colonial British government announced in 1865 that forests were transformed into state property (Guha, 1983).

After independence, the Indian government did not reverse the main centralization trend of colonial forest policy, but rather strengthened it through a series of legislative and other measures enhancing government's control over forest resources and multiplying restrictions imposed upon the tribal population in view of continuous forest degradation (Baland and Platteau, 1996). Since independence, there have been three forest policy pronouncements in India: the 1952 Forest Policy, the National Commission on Agriculture (NCA) 1976, and the 1988 Forest Policy (Saxena, 1999). It is widely argued that the four decades of forest policy preceding 1988 were mainly concerned with timber production for commercial purposes, and have neither been sustainable in terms of checking the process of deforestation, nor have they improved people's access to forests for meeting their basic needs.

In the light of massive deforestation and forest degradation leading to increased social unrest and conflict between state and village communities, the 1988 Forest Policy was a landmark in Indian forest policy. It implied a radical shift from the earlier revenue orientation towards conservation now being considered a priority. Commercial forest exploitation by industries became prohibited, while soil conservation and other ecological functions as well as subsistence requirements of the local people became major objectives. In a follow-up document issued in 1990, the central government issued guidelines to all state governments to implement 'Joint Forest Management Systems' in order to regenerate degraded forest and improve living standards (GoI, 1990). The guidelines suggested that the state governments might devolve everyday forest protection, management and development responsibilities to local institutions at the village level (Forest Protection Committees (FPCs)) and prescribe benefit-sharing arrangements following regeneration, implying a clear shift from a system of pure

state management to one of co-management. In 2000, a revised JFM resolution was issued, containing some significant changes, e.g., the extension of JFM to good forests and the requirement that 33% of FPC members be women (GoI, 2000). Within the broad framework of rules and regulations provided by the central government, the state governments have scope to implement their own resolutions of rules and regulations, depending on their prevailing requirements and situations. As we will see below, state resolutions vary widely. Little is known about the underlying reasons for this observed inter-state asymmetry in JFM implementation. Certainly, the interests of FDs and the governing political party influence the decisions. Donor activities also play an important role (Damodaran and Engel, 2003). Decisions are rarely made on the basis of economic, social and institutional efficiency, but rather respond to pressures from various stakeholder groups (Khare et al., 2000; Saxena, 1999; Vira, 1995).

Is the shift in the overall property rights regime from one of state management to co-management desirable? Bulte and Engel (in press) point out how the state and local communities differ in their abilities to deal with different sources of market failure that are at the heart of the overexploitation of forests. The state generally has an advantage in dealing with issues of externalities, public goods, and scale which are an important source of market failure in forest management. On the other hand, high transaction costs associated with monitoring and enforcement of state-imposed use and access rules and a lack of information about local conditions and needs have been major reasons behind the government's failure to effectively protect forests in the past. The informal sanctioning mechanisms, repeated interactions, and better access to local information within communities can save on such transaction costs (McKean, 2000). Moreover, the lack of formalization of customary rights under state management implies tenure insecurity on the part of traditional forest-dependent communities. This is a potential advantage of devolving formal rights to local communities. Communities and the state also differ in their ability to deal with policy failure (Bulte and Engel, in press). While state management has often tolerated illegal logging activities in order to relieve social pressures or extract rents from commercial logging, empirical

evidence also indicates that community elites may manipulate devolution outcomes in their own interest, at the cost of weaker segments of the communities (Shackleton et al., 2002; Platteau, 2003).

It is clear that all of these considerations are applicable to the case of forest management in India. Thus, both communities and the state have potential strengths and weaknesses in forest management, indicating that the move towards a co-management regime such as JFM is certainly a step in the right direction (Bulte and Engel, in press). At least three necessary conditions for effective co-management emerge from a review of the literature (Ribot, 2002; Larson and Ribot, 2004): (i) secure and well-defined property rights; (ii) transfer of appropriate and sufficient powers to communities, and (iii) downwardly accountable and representative local institutions. It will be shown in other parts of this paper that in the case of India these conditions have not been met. Rather, the devolution process has often been incoherent and incomplete and has not fully exploited the comparative strengths and weaknesses of the two parties involved.

3.2. Degree of property rights transfer

To what degree have specific types of property rights been devolved from the state to local communities (FPCs) under JFM? In this chapter we focus on formal rules. This includes national-level JFM guidelines and state-level regulations. It should be noted, however, that JFM is based on administrative orders. As such, JFM rules only become formally binding once an FPC signs up to the JFM program. The latter constitutes a contract between individual FPCs and the corresponding FD, with differences in individual rules and their enforcement (see Section 4).² In this section we distinguish four types of property rights that are most relevant for the case of common-pool resources such as forests (Schlager and Ostrom, 1992): (1) the right to enter a defined physical area and obtain resource units or products of a resource system (e.g., extracting timber and NTFPs) (*withdra-*

wal right), (2) the right to regulate internal use patterns and transform the resource by making improvements (e.g., planting seedlings and thinning trees) (*management right*), (3) the right to determine who will have an access right, and how that right may be transferred (*exclusion right*), (4) the right to sell or lease management and exclusion rights (*alienation right*).

In general, the degree to which specific rights and benefits were actually devolved from the FD to local communities differs significantly across states (Damodaran and Engel, 2003). In 12 out of 23 states that have implemented JFM, the FD has unilateral power to cancel the JFM agreement and in most cases to even dissolve the FPC (Khare et al., 2000). Most state program guidelines do not provide long-term rights to communities (Poffenberger and McGean, 1996). The legal status of FPCs also differs between states. In general, FPCs are not legal bodies unless they are registered under the Societies Act or Co-operatives Act. Such registration is only mandatory in 6 states, and optional in two states (Ravindranath et al., 2000).

3.2.1. Withdrawal rights

Under JFM, members of FPCs have obtained the right to use several non-timber forest products from their village forests, and to keep a share of the proceeds from the sale of timber once forests have matured. However, withdrawal rights are not granted to communities for all NTFPs. In particular, the right to collect *beedi*, *sal* and *tendu* leaves, which constitute lucrative forest produce for the governments and contractors, has not been fully devolved to communities. For instance, in Andhra Pradesh a total of 22 NTFPs are given in lease to a state government agency called GCC (Girijan Cooperative Corporation), thereby prohibiting collectors from the villages to sell these products in the open market, despite the fact that tribal people's livelihood depends on many of these NTFPs.³ Table 1 presents response statistics from our field survey for those households that were

² As such the interpretation of official guidelines and provisions as formal or absolute property rights is a bit artificial, but is made here in order to structure the analysis. We would like to thank an anonymous reviewer for highlighting this point.

³ According to the Andhra Pradesh Schedule Areas Minor Forest Produce (MFP) Regulation (of trade) Act of 1979, GCC is the monopoly agent of the government of Andhra Pradesh to purchase NTFPs in schedule areas. NTFP trade outside schedule areas is vested with the FD as per the Andhra Pradesh Minor Forest Produce Regulation (of trade) Act 1971. However, the FD also chose GCC to entrust NTFP collection in non-schedule areas (CPF, 2003).

Table 1
Household's responses regarding specific restrictions on NTFP collection and sales

Type of restrictions	Percentage
Sell only to contractor	29.8
Sell only to government depot	62.9
Not allowed to collect	7.2

Source: Field survey ($n=229$).

involved in collection of NTFPs in Andhra Pradesh. As can be seen from the table, about 63% of households reported that they sell their products to a government depot, and approximately 30% reported selling to contractors.

Villagers also do not have the right to harvest timber products, unless the wood is dry or has fallen to the ground. In some cases, their share of the proceeds from government timber harvests is limited, and has been reduced after successful protection (Agrawal and Ostrom, 2002). The share of forest benefits obtained by communities varies across states and between types of products, ranging from 0% (for timber in Bihar) to 100% (e.g., for NTFPs in Andhra Pradesh) (SPWD, 1998). In several states it is also mandated that a certain percentage of forest benefits is to be used in a village development fund (ibid).

Finally, a very important restriction in the transfer of withdrawal rights according to national JFM guidelines is that, in the event that FPCs fail to protect the forests, the government can withdraw the usufruct rights from the communities unilaterally without any compensation for previous efforts of forest regeneration (GoI, 1990).

3.2.2. Management rights

The right to manage forests is partially transferred to the FPCs under JFM. The communities have the right to select tree species to plant, but only subject to approval by the FD. A joint working plan or micro plans outlining a detailed management plan are to be prepared by the FD in consultation with the FPCs. In some cases, such as in Andhra Pradesh, the preparation of the micro plan is entrusted to the local communities, but the FD needs to give final approval of the plan to implement it. According to TERI (2004), in 9 out of 22 states, the FD retained the right to frame rules, with communities either having no rights (in 5 states) or the right to assist the FD (in 4 states). The

distribution of benefits from the forest is another important management right. However, 7 states did not transfer this right to communities, and in another 6 states the communities only had the right to assist the FD (TERI (2004)).

3.2.3. Exclusion rights

Under JFM, the right to determine who will have access rights is formally granted to the FPCs. According to national JFM guidelines, it is the communities who should be able to organize themselves into a village institution for the protection and regeneration of the forests and become beneficiaries from it (GoI, 1990). This implies that the communities have the right to decide who has access to the forest and who has not. Although a few communities have succeeded in excluding non-members, the communities' ability to enforce access rights is often limited. Only in 7 out of 22 states for which information was available, communities had the right to punish violators of rules by charging a fine. In 7 states they could only assist the FD in this regard, while in another 7 states they did not have any such right (TERI, 2004).⁴ Another important exclusion right and potential means of rule enforcement is the community's right to cancel an individual household's membership. In 12 states the communities did not have such right, and in another 2 states they could only assist the FD (TERI (2004)).

3.2.4. Alienation rights

Alienation rights, i.e., the right to sell or transform a resource, are considered by the property rights school as necessary to assure an efficient resource use. Under JFM, these rights have been transferred to communities only to a very limited degree. It is very clearly mentioned in the guidelines that no ownership or lease rights over forest land shall be given to communities (GoI, 1990). However, the other rights (withdrawal, management and exclusion) over forests can be transferred by the community from one generation to another. The right to transform the resources is also limited to afforestation activities.

To conclude, it appears that the conditions for effective co-management—particularly the conditions

⁴ This information was generated from the individual state JFM guidelines available at the TERI website.

that property rights should be secure and well-defined, and that appropriate and sufficient powers be transferred to communities (Ribot, 2002; Larson and Ribot, 2004)—are not satisfied in the case of JFM. Rights are insecure because they are only administrative, not legal rights, and as such can be withdrawn at any time. The transfer of property rights to communities has been partial, and its degree differs widely across states. While it may be conceptually appropriate that the state retains some rights, e.g., the determination of forest areas to be included to account for externalities, it appears that it often also retains rights where the community would conceptually have an advantage (e.g. the right to punish and cancel membership). The incomplete transfer of rights is likely to result in reduced community incentives for sustainable resource management and provides incentives to over-exploit forests for quick economic benefits (Bulte and Engel, *in press*).

4. Governance of contractual relations (play of the game)

In this chapter we examine level 3, i.e., the governance of contractual relations (contracts and enforcement). Here we take a closer look at the play of the game in terms of interactions between individual FPCs and FDs. The fact that formal rights are administrative rights only, leaves a lot of room for variation in actual contractual relations. Thus, this level is concerned with how formal rules are translated or enforced and its implications for different outcomes. The need for a good governance structure arises particularly when there are unforeseen problems in the contractual obligations in which parties are involved (Williamson, 2000). Such problems arise mainly due to two reasons: (1) lack of foresight, and (2) asymmetric information (Furubotn and Richter, 2000).

Lack of foresight refers to a situation where parties are unable to foresee precisely the future consequences of the contract. This problem can be tackled effectively if the contractual obligations are adaptable to new and unexpected situations. This is certainly an issue in JFM, but unfortunately no specific information was available. The problem of asymmetric information arises when one party knows more than the

other in regard to the information related to the state of contract. This may lead to opportunism for the party who has the informational advantage and may create serious problems of contractual performance.

In what follows, we first discuss the rule setting process under JFM at the local level. Second, we analyze problems of rule enforcement. And thirdly we highlight problems of asymmetric information and rent-seeking between the FD and local communities with the help of agency theory.

4.1. Rule setting

The process of rule setting is an important issue in natural resource management in general and for JFM in particular. Rule setting in JFM is the result of complex procedures as there are layers of hierarchies in bureaucratic and government agencies involved. State government guidelines make up the basic framework of rules for FDs, which in turn lay down rules for FPCs governing the use and management of forest resources. FPC members are responsible for harvesting, distributing benefits, and regulating access to forest produce according to these rules. Since JFM is a co-management system, not all authorities for making constitutional, collective choice and operational level rules are concentrated in the FPCs. The FD retains its authority to make rules at three levels: (i) area allocation and boundary demarcation; (ii) preparation and approval of micro plans and designation of forest protection methods; and, (iii) determination of disposal procedures for commercially valuable NTFPs (Prasad, 2002; Saxena, 1997). The provision in most state resolutions that a forest officer be an FPC member indicates the influence of FDs on community level decision-making. Studies show that, in practice, FPCs are often completely dependent on FDs for day-to-day activities and decision-making processes (Ballabh et al., 2002; Sarin et al., 1998; Saxena, 1997). Moreover, micro plans are frequently prepared by the FD or NGO personnel with little consultation of the FPCs (Rangachari and Mukherji, 2000). The degree of FD's intervention in the FPCs varies, however, from state to state (Kumar, 2000; Sarin, 2002; Ballabh et al., 2002). In short, FD officials significantly shape the degree to which villagers can frame their own rules of use and management, and get involved

in day-to-day management of village forests (Jeffery and Sunder, 1999).

Our field experience in Andhra Pradesh showed communities being instructed by the FD on what activities they should undertake in their forest areas. It was found that communities have invariably adopted silvicultural procedures such as singling and coppicing to raise good quality timber irrespective of the tree species. As a result, a drastic decline in NTFPs was reported. Plantation activities are frequently carried out against the wishes of large segments of the community. Communities also frequently have to await permission from the FD officials before taking an initiative. Table 2 presents the perceptions of households interviewed in our field study regarding the FD in the process of forest management under JFM. It shows that nearly 20% of the households feel that the attitude of the FD officials towards the people was dominating. Villagers have the responsibility to protect trees and vegetation and the ability to allocate non-timber benefits within the community. But where more significant issues are at stake such as transfer of power to villagers, formation of FPCs, share in timber sales, or negotiating rule changes, villagers are highly constrained in their ability to act (Kumar, 2000).

Governments also often maintain control through extensive bureaucratic procedures (forest management plans), withholding information and lack of capacity building (e.g. in the state of Madhya Pradesh locals were completely unaware of rules for budgetary allocations or use of funds) (Sundar, 2001).

4.2. Enforcement of rules

The basic purpose of institutions is to structure human behavior in order to achieve the settled objec-

tives. If the objectives are set and rules are framed to achieve them, an effective mechanism must be implemented to make sure that the rules are enforced and observed. Effective enforcement requires a mechanism to detect rule violations and proper sanctions to deter such violations.

FPCs and FD are, through various contractual relations, bound to certain rules that concern benefit sharing, forest protection and management. Violation of such rules from both sides is, however, common. Communities violate rules when they cannot protect the forests from fire, grazing and tree felling. FDs violate rules when they indulge in illegal activities in forests protected by FPCs, e.g., by supporting unauthorized persons (smugglers, grazers, contractors, and headloaders) in using the forests for their own rent-seeking interests (see Section 4.3) or by failing to provide required administrative and legal support to the FPCs in resolving conflicts with outsiders and neighboring villages. Many such instances of violation committed by the FD staff were reported during the implementation process of the JFM program and also later on (own observation).⁵

No detection and sanctioning mechanisms are framed in the JFM guidelines that deter the FD officials from violating rules, whereas enforcement mechanisms exist for the event that FPCs commit violations. The guidelines stipulate that if the communities fail to protect the forests, all rights to use and manage the forests will be withdrawn without any compensation for previous efforts towards forest regeneration. In reality, however, no community has been reported punished for their negligence. Rather it is often seen that whenever communities are disintegrated and commit large-scale violations in the forests, FDs tend to spend more money on forest activities. This not only facilitates rent seeking by the FD in the event of lack of monitoring from the communities, but also encourages the FPCs not to take rules seriously. Due to lack of deterrence

Table 2

Household perception in the evaluation of the attitude of the Forest Department in the whole process of forest management under JFM (in percentage)

Attitude of the FD	Percentage
Very dominating	1.51
Dominating	17.60
Very cooperating	5.71
Cooperative	56.86
Cannot say	18.31
Total	100

Source: Field survey ($n=643$).

⁵ In the majority of the 147 villages visited problems of illegal activities by headloaders and grazers were reported. These headloaders and grazers feel that their action is legitimate since they make regular payments to the local FD officials informally for access to and use of forests (own fieldwork).

mechanisms, rampant rent-seeking activities are reported in Andhra Pradesh (see Section 4.3).⁶

Moreover, the communities' ability to enforce access rights and other rules among community members is often limited as it depends on the FD's willingness to provide support in terms of administrative and legal backing and conflict resolution between FPC members and non-member forest users. For instance, in a meeting with JFM presidents observed during our fieldwork, a FD official in charge of JFM announced that the committees should allow access to forests by grazers. In a few instances, the FD forced the committees to accommodate non-members against FPC wishes. This kind of erratic change of policy on the part of FDs violates the rights of exclusion given to the communities. The administrative nature of JFM constrains the ability of FPC members to take any legal measures to protect their rights. Moreover, it is in the interest of the FD to ensure that JFM does not work properly because once a community institution becomes vibrant, it is more likely to oppose the illegal access to their forests by outsiders.

4.3. Problems of asymmetric information and rent-seeking

Problems of asymmetric information and rent-seeking are widespread under JFM. The relationship between FD and local communities in JFM institutions can be analyzed with the help of agency theory as a principal–agent problem. The core of agency theory is that there is a hierarchical system, in which a principal delegates responsibilities to an agent, who due to asymmetries of information and costliness of monitoring has a significant discretionary decision margin. In a democratic society like India, the people (citizens) can be considered as principals electing representatives (to the central and state governments) to rule on their behalf (Bardhan, 2002). Furthermore, the agents' (governments') wishes are implemented by civil servants (here, FD bureaucrats), who are sub-agents and have their own interests. The bureaucrats are

not elected representatives and therefore are not accountable to the people. Instead, they are accountable to their seniors in the civil service and enjoy security of tenure and protection by the Constitution (Vyasulu, 2004). Thus, FD bureaucrats have their own interests in managing forests in a way that suits the timber and revenue needs of the government (silvicultural methods) and their own interests of rent seeking. Due to a lack of accountability, often combined with an ignorance of local communities' knowledge, customs, and habits and the uplifting feeling of being educated in a largely illiterate society, these bureaucrats often think themselves more knowledgeable than the local communities. The source of the asymmetric information problem is that FDs usually have better information than communities about the details of JFM provisions, costs and benefits of individual forest activities (e.g., equipment, timber revenues), as well as potential loopholes of the JFM contract. As a consequence, FD officials considerably limit access to information for local communities regarding allocation of funds and other related matters, yielding opportunities for rent-seeking (Rangachari and Mukherji, 2000).

Several types of FD rent-seeking could be observed during our fieldwork. First, out of the 55 communities interviewed, 49% reported that bureaucrats retain a percentage share of the allocated funds intended for carrying out forest activities. Second, 35% of these villages reported mismanagement by FD officials of funds allocated for village development, and another 15% indicated misuse of funds earmarked for capacity building such as training and exposure visits. Third, the lack of transparency in the transaction proceedings of harvested timber and NTFPs often leaves communities at the mercy of the FD as regards revenue sharing from harvests.⁷ Out of

⁶ Informal discussions with presidents, NGOs, and few FD officials revealed widespread rent seeking activities by FD officials in all three districts sampled (own fieldwork).

⁷ The lack of transparency arises as FDs keep the harvested timber in their timber depots before negotiating with potential buyers, and the negotiation process does not involve FPC leaders. In all, except one, community where timber and bamboo were harvested for sales, the entire harvesting activities (from logging to final sale) were carried out by the FD with little and sometimes no participation of local communities. The communities had no idea about the rate at which their products were sold and the amount of money received. As a consequence, the FD often claimed that large amounts were spent in transportation and other expenses without providing details of proceedings. While communities often thought that FD payments were unreasonable, they had no way of measuring actual benefits.

eight communities that had already harvested timber or bamboo for sales, seven reported rent-seeking in the form of FDs misreporting costs or revenues from the operations. It should be noted that these results are likely to underestimate the true extent of rent-seeking as many communities were reluctant to discuss the issue.

Furthermore, as stated earlier, the lower cadres of the FD staff, in charge of the implementation of the JFM program, have an incentive to undermine monitoring and enforcement of use and access rules in return for bribes.

Moreover, rent-seeking by FD officials is often facilitated by the lack of accountability of FPC leaders to the members. It is important to note that most forest-dependent communities are marginalized members of Indian society who live under abject poverty and are vulnerable to the exploitation of local and national elites and this lowers their ability to enforce any contracts. The role of FPC leaders under this circumstance is very crucial in defending the rights of their members. The FD, however, often directly influences the election of FPC leaders by proposing the candidates to select from. As a result, the leaders who get elected consider themselves obliged to the FD and dependent on it for reelection. In many cases, leaders have been occupying the posts since the formation of FPCs, despite member requests for a change and the JFM resolution allowing the election of presidents and secretaries every two years. Having a close relation with the FD, the FPC leaders have failed to protect the contractual rights of the marginalized groups whom they represent.

5. Concluding remarks

In this paper we have analyzed the performance of JFM and the sources of persistent problems by structuring the evidence according to Williamson's four levels of institutional analysis and focusing particularly on levels 2 and 3.

The analysis of the 'rules of the game' (level 2) has highlighted the fact that—while the move towards co-management is certainly a step in the right direction—the transfer of property rights from the FD to local communities has been incomplete and incoherent and continues to be subject to much uncertainty. As a

consequence, FPCs face serious disincentives to consider the full long-run effects of their actions. Transforming administrative rights into legal rights would help to improve the security of community rights. Registration of FPCs as cooperatives is another approach that can provide a more limited extent of security. The more consistent transfer of rights to communities in areas where the community conceptually has an advantage, e.g. the rights to punish and cancel membership, would be desirable to increase the efficiency of the co-management system. In developing strategies for achieving such institutional change, a better understanding of the underlying processes and factors that have led to the observed asymmetry in implementation across states is required.

The third-level analysis has revealed severe problems in the 'play of the game', including serious enforcement and rent-seeking problems. It was shown that the FD severely limits and often even undermines the scope for the community to set and enforce effective use and access rules. It not only routinely intervenes in everyday decision-making within the community. Rent-seeking activities also lead to FD officials supporting the violation of FPC rules. The lack of accountability and the prevalence of information asymmetries of both FPC leaders as well as FD bureaucrats is an important factor in explaining the strong prevalence of rent-seeking activities. An independent vigilance system to monitor and supervise JFM activities at the village level may be one potential approach to improving accountability. The involvement of existing institutions, such as the *panchayats* as well as the identification and engagement of reputed and committed NGOs would be other approaches. The results also point to the importance of capacity building and information provision. Again, communities should also be empowered with legal backing so that they can take cases of abuse directly to court.

Because individual levels are inter-connected and problems on one level can have tremendous repercussions on the other levels, policy formulation needs to focus on all levels. It should thus be stressed that levels 1 (informal institutions) and 4 (household incentives for participation) are likely to be equally important for understanding the evolution of JFM and for developing strategies to overcome current inefficiencies. An analysis of these levels and an integration with the present study is an important venue for future research.

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References

- Agrawal, A., Ostrom, E., 2002. Collective action, property rights, and decentralization in resource use in India and Nepal. *Politics and Society* 29 (4), 485–514.
- Ballabh, V., Balooni, K., Dave, S., 2002. Why local resources management institutions decline: a comparative analysis of Van (forest) panchayats and forest protection committees in India. *World Development* 30 (12), 2153–2167.
- Baland, J-M., Platteau, J.P., 1996. *Halting Degradation of Natural Resources: Is there a Role for Rural Communities?* Clarendon Press, Oxford.
- Bardhan, P., 2002. Decentralization of governance and development. *Journal of Economic Perspectives* 16 (4), 185–205.
- Bulte, E., und Engel, S., in press. Conservation of tropical forests: Addressing market failure. http://www0.gsb.columbia.edu/ipd/pub/ConservationofTropicalForests11_29_04.pdf. In: López, R., Stiglitz, J., Toman, M. (Eds.), *Sustainable development: New options and policies*. Oxford University Press, New York.
- CPF, 2003. VSS Sustainability and the role of GCC (NTFP marketing) in connection with CFM in Andhra Pradesh. Action research series 1, Centre for People's Forestry, Secunderabad, Andhra Pradesh, India.
- Damodaran, A., Engel, S., 2003. Joint forest management in India: assessment of performance and evaluation of impacts. ZEF-discussion papers on development policy 77. Center for Development Research, Bonn.
- Deaton, Angus S., 1997. *The Analysis of Household Surveys: A Microeconometric Approach to Development Policy*. Johns Hopkins University Press, World Bank, Washington, DC.
- Demsetz, H., 1967. Towards a theory of property rights. *American Economic Review; Papers and Proceedings* 57, 347–359.
- Furubotn, E.G., Richter, R., 2000. *Institutions and Economic Theory*. University of Michigan Press, Ann Arbor.
- Ghate, R., 2003. Ensuring collective action in participatory forest management. South Asian Network for Development and Environmental Economics, Nepal. Working Paper 3-03.
- GoI, 1990. Involving of village communities and voluntary agencies for regeneration of degraded forest lands (Letter no. 6-21/89-PP, June 1, 1990). Government of India, Ministry of Environment and Forests, New Delhi.
- GoI, 2000. Guidelines for strengthening of joint forest management program (Letter no. 22-8/2000-JFM (FPD), February 21, 2000). Government of India, Ministry of Environment and Forests, New Delhi.
- Guha, R., 1983. Forestry in British and Post British India: A Historical Analysis, Economic and Political weekly XXVIII 44.
- Hardin, G., 1968. The tragedy of the commons. *Science* 162, 1243–1248.
- Jeffery, R., Sunder, N., 1999. *A New Moral Economy for India's Forests? Discourses of Community and Participation*. Sage Publication, New Delhi.
- Joshi, A., 1999. Progressive Bureaucracy: An Oxymoron? The case of joint forest management in India. Rural development forestry network. Oversea Development Institute, London.
- Khare, A., Sarin, M., Saxena, N.C., Palit, S., Bathla, S., Vania, F., Satyanarayana, M., 2000. Joint forest management: Policy, practice and prospects. Policy that works for forest and people 3. World Wide Fund for Nature, India, and IIED, London.
- Kumar, N., 2000a. All is not green with JFM in India, in forest, trees and people, Newsletter No. 42, June, Rome.
- Kumar, S., 2002b. Does participation in common pool resource management help the poor? A social cost-benefit analysis of joint forest management in Jharkhand, India. *World Development* 30 (5), 763–782.
- Larson, A.M., Ribot, J.C., 2004. Democratic decentralization through a natural resource lens: an introduction. *The European Journal of Development Research* 16 (1), 1–25.
- McKean, M.A., 2000. Common property: what is it, what is it good for, and what makes it work? In: Clark, C., McKean, M., Ostrom, E. (Eds.), *People and forests: Communities, Institutions and Governance*. MIT Press, Cambridge, Mass., pp. 29–51.
- North, D.C., 1990. *Institutions, Institutional Change and Economic Performance*. Cambridge University Press, Cambridge.
- Ostrom, E., 1990. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, Cambridge.
- Panayotou, T., 1993. *Green Market: The Economics of Sustainable Development*. Institute for Contemporary Studies Sans Francisco, California.
- Pearce, D.W., Warford, J.J., 1993. *World Without End: Economics, Environment and Sustainable Development*. Oxford University Press, New York.
- Platteau, J.-P., 2003. Decentralized development as a strategy to reduce poverty? Centre de Recherche en Economie de Développement (CRED), University of Namur, Belgium.
- Poffenberger, M., McGean, B., 1996. *Village Voices, Forest Choices: Joint Forest Management in India*. Oxford University Press, New Delhi.
- Prasad, R., 2002. Joint forest management in India and the impact of sate control over non-wood forest products. Indian Institute of Forest Management, Bhopal.
- Rangachari, C.S., Mukherji, S.D., 2000. *Old Roots, New Shoots: A Study of Joint Forest Management in Andhra Pradesh*. Winrock International, New Delhi.
- Ravindranath, N.H., Murali, K.S., Malhotra, K.C. (Eds.), 2000. *Joint forest management and community forestry in India. An ecological and institutional assessment*. Oxford and IBH Publishing.
- Ribot, J., 2002. Democratic decentralization of natural resources, institutionalizing popular participation. World Resources Institute, Washington D.C.

- Sarin, M., 2002. Devolution as a threat to democratic decision-making in forestry? Findings from three states in India. CIFOR, Indonesia.
- Sarin, M., Ray, L., Raju, M.S., Chatterjee, M., Banerjee, N., Hir-emath, S., 1998. Gender and equity concerns in joint forest management. In: Kothari, M., et al., (Eds.), *Communities and Conservation: Natural Resource Management in South and Central Asia*. Sage Publication, New Delhi.
- Saxena, N.C., 1997. The saga of participatory forest management in India. CIFOR, Indonesia.
- Saxena, N.C., 1999. Forest policy in India, policy and joint forest management series 1. World Wide Fund for Nature, India.
- Schlager, E., Ostrom, E., 1992. Property rights regimes and natural resources: a conceptual analysis. *Land Economics* 68 (3), 249–262.
- Shackleton, S., Campbell, B., Wollenberg, E., Edmunds, D., 2002. Devolution and community-based natural resource management: creating space for local people to participate and benefit. *Natural Resource Perspectives*, vol. 76. Oversea Development Institute, London.
- SPWD, 1998. JFM Status Book. Society for Promotion of Waste-land Development, New Delhi.
- Sundar, N., 2001. Is devolution democratization? *World Development* 29 (12), 2007–2023.
- TERI, 2004. Joint forest management. Tata Energy Research Institute, India. www.teriin.org/jfm/issues/benefit.htm (accessed 28/1/2004).
- Tietenberg, T., 1994. *Environmental Economics and Policy*. Harper Collins College, New York.
- Vira, B., 1995. Institutional change in India's forest sector, 1976–1994: reflections on state policy, OCEES Research Paper, vol. 5. London.
- Vyasulu, V., 2004. Transformation in governance since 1990s: some reflection. *Economic and Political Weekly* XXXIX (23), 2377–2385.
- Williamson, O., 2000. The new institutional economics: taking stock, looking ahead. *Journal of Economic Literature* XXXVIII, 595–613.